

# Waves

## SHORT NOTES

### General Equation of Wave Motion

$$\frac{d^2 y}{dt^2} = v^2 \frac{d^2 y}{dx^2}$$

$$y(x, t) = f\left(t \pm \frac{x}{v}\right)$$

where,  $y(x, t)$  should be finite everywhere.

$f\left(t + \frac{x}{v}\right)$  represents wave travelling in -ve x-axis.

$f\left(t - \frac{x}{v}\right)$  represents wave travelling in + ve x-axis.

$$y = A \sin(\omega t \pm kx + \phi)$$

### Terms Related to Wave Motion (For 1-D Progressive Sine Wave)

#### Wave Number (or Propagation Constant) (k)

$$k = 2\pi / \lambda = \frac{\omega}{v} \text{ (rad m}^{-1}\text{)}$$

#### Phase of Wave

The argument of harmonic function  $(\omega t \pm kx + \phi)$  is called phase of the wave.

Phase difference ( $\Delta\phi$ ): difference in phases of two particles at any time  $t$ .

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta x \text{ where } \Delta x \text{ is path difference}$$

$$\text{Also } \Delta\phi = \frac{2\pi}{T} \Delta t$$

### Speed of Transverse Wave Along the String

$$v = \sqrt{\frac{T}{\mu}} \text{ where } T = \text{Tension}$$

$\mu$  = mass per unit length

### Velocity of Longitudinal Waves

- ❖ Velocity of longitudinal waves in solid,  $v = \sqrt{\frac{Y}{\rho}}$
- ❖ Velocity of longitudinal waves in liquid and gas,  $v = \sqrt{\frac{K}{\rho}}$   
where,  $Y \rightarrow$  Young's modulus,  $K \rightarrow$  Bulk modulus.

#### Newton's Formula:

$$\text{Velocity of sound in gas, } v = \sqrt{\frac{P}{\rho}}$$

#### Laplace Formula:

$$v = \sqrt{\frac{\gamma P}{\rho}}, \text{ where } \gamma = \frac{C_P}{C_V} \text{ and } P = \text{adiabatic pressure.}$$

### Power Transmitted Along the String

$$\text{Average Power } \langle P \rangle = 2\pi^2 f^2 A^2 \mu v$$

$$\text{Intensity } I = \frac{\langle P \rangle}{s} = 2\pi^2 f^2 A^2 \rho v$$

### Reflection of waves

If we have a wave

$$y_i(x, t) = a \sin(kx - \omega t) \text{ then,}$$

(i) Equation of wave reflected at a rigid boundary

$$y_r(x, t) = a \sin(kx + \omega t + \pi)$$

$$\text{or } y_r(x, t) = -a \sin(kx + \omega t)$$

i.e. the reflected wave is  $180^\circ$  out of phase.

(ii) Equation of wave reflected at an open boundary

$$y_r(x, t) = a \sin(kx + \omega t)$$

i.e. the reflected wave is in phase with the incident wave.

### Standing/Stationary Waves

$$y_1 = A \sin(\omega t - kx + \theta_1)$$

$$y_2 = A \sin(\omega t - kx + \theta_2)$$

$$y_1 + y_2 = 2A \cos\left(kx + \frac{\theta_2 - \theta_1}{2}\right) \sin\left(\omega t + \frac{\theta_1 + \theta_2}{2}\right)$$

The quantity  $2A \cos\left(kx + \frac{\theta_2 - \theta_1}{2}\right)$  represents resultant amplitude at  $x$ . At some position resultant amplitude is zero these are called **nodes**. At some positions resultant amplitude is  $2A$ , these are called **antinodes**.

$$\text{Distance between successive nodes or antinodes} = \frac{\lambda}{2}$$

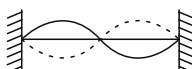
$$\text{Distance between adjacent nodes and antinodes} = \lambda/4.$$

All the particles in same segment (portion between two successive nodes) vibrate in same phase.

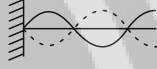
Since nodes are permanently at rest so energy can not be transmitted across these.

## Vibrations of Strings (Standing Wave)

### Fixed at Both Ends

|                                                           |                                                                   |                                                                                     |
|-----------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| First harmonics or Fundamental frequency                  | $L = \frac{\lambda}{2}, f_1 = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$  |  |
| Second harmonics or First overtone                        | $L = \frac{2\lambda}{2}, f_2 = \frac{2}{2L} \sqrt{\frac{T}{\mu}}$ |  |
| Third harmonics or Second overtone                        | $L = \frac{3\lambda}{2}, f_3 = \frac{3}{2L} \sqrt{\frac{T}{\mu}}$ |  |
| $n^{\text{th}}$ harmonics or $(n-1)^{\text{th}}$ overtone | $L = \frac{n\lambda}{2}, f_n = \frac{n}{2L} \sqrt{\frac{T}{\mu}}$ |  |

### String Free at One End

|                                                           |                                                                                  |                                                                                       |
|-----------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| First harmonics or Fundamental frequency                  | $L = \frac{\lambda}{4}, f_1 = \frac{1}{4L} \sqrt{\frac{T}{\mu}}$                 |    |
| Third harmonics or First overtone                         | $L = \frac{3\lambda}{4}, f_3 = \frac{3}{4L} \sqrt{\frac{T}{\mu}}$                |    |
| Fifth harmonics or Second overtone                        | $L = \frac{5\lambda}{4}, f_5 = \frac{5}{4L} \sqrt{\frac{T}{\mu}}$                |    |
| $(2n+1)^{\text{th}}$ harmonic or $n^{\text{th}}$ overtone | $L = \frac{(2n+1)\lambda}{4}, f_{2n+1} = \frac{(2n+1)}{4L} \sqrt{\frac{T}{\mu}}$ |  |

## Organ Pipes

- In a closed organ pipe only odd harmonics are present.

$$v_1 = \frac{V}{4L} \quad (\text{fundamental})$$

$$v_2 = 3v \quad (\text{third harmonic or first overtone})$$

$$v_3 = 5v$$

$$v_n = (2n-1)v$$

- In an open organ pipe both odd and even harmonics are present.

$$v'_1 = \frac{V}{2L} = v' \quad (\text{first harmonic})$$

$$v'_2 = 2v' \quad (\text{second harmonic or first overtone})$$

$$v'_3 = 3v'$$

$$v'_n = (2n-1)v'$$

- Resonance tube: If  $l_1$  and  $l_2$  are the first and second resonance length with a tuning fork of frequency ' $v$ ' then the speed of sound.

$$v = 4v(L_2 + 0.3D)$$

Where,  $D$  = internal diameter of resonance tube

$$v = 2v(l_2 + l_1)$$

$$\text{End correction} = 0.3D = \frac{l_2 - l_1}{2}$$

## Beats Frequency

- ❖ Beat frequency = Difference in frequency of two sources = No. of beats per second.

$$\text{beat} = v_1 - v_2$$

$$v_2 = v_1 \pm \text{beat}$$

- ❖ Beat frequency is always a positive value. This fact can be used to decide about + or – sign in the above equation.